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Policies, incentives, and faculty participation in the Institutional sustainability of public higher education in Panama

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This qualitative study analyzes determining factors in the achievement of scientific and academic research in Panama's public universities through methodological and data triangulation. The research explored academics'perceptions about the research ecosystem considering data from a national survey (ENACT; *n* = 565 responses analyzed), six semi-structured interviews, and an exploratory post-seminar perception poll completed by 117 academics. Findings indicate important advances in policies, incentives, and support structures for research. However, challenges persist: sustained budgetary limitations, complex bureaucratic processes, and cultural barriers that hinder the optimization of research potential. A perceptual gap was identified between university authorities and the academic community; while authorities emphasize normative and structural advances, academics express doubts about the continuity and effectiveness of these measures. The study also identified an imbalance between the growing international visibility of scientific production and its limited impact on local problems and public policy. Results suggest the need to implement strategies that combine: strengthening university governance with active academic participation, guaranteeing sustained funding through state-university-private sector articulation, diversifying incentives, and reducing bureaucratic barriers. It is recommended to enhance scientific communication and orient research agendas with the purpose of contributing to global knowledge while responding to national development priorities, consolidating university research as an engine of the country's socioeconomic and cultural development.

KEYWORDS
faculty participation, higher education, research barriers, research culture, research public policies, scientific research, institutional sustainability

1 Introduction

In recent decades, scientific research has taken on an increasingly central role in public universities in Panama, reflecting a global trend toward integrating research as a fundamental pillar of higher education and universities as social actors that promote it (García-Arce et al., 2021). Barsallo et al. (2025) posit that factors such as intellectual stimulation and social contribution are currently the most motivating factors for re-search activity among

academics at Panamanian public universities. Consequently, research has been driven by the growing need to generate knowledge applicable to local and global challenges, as well as by the demand for constant innovation in all fields of knowledge. The Mendigutxia (2024), highlights that Latin American universities generate more than 80% of regional scientific output, underscoring their role as pillars of scientific development. Gutiérrez-Sánchez et al. (2025) and Huete-Perez and Salvatierra (2025) support this and add that although it is not a substantial contribution as compared with other countries, Panama exhibits an emerging but modest research activity. Nevertheless, the improvement of research capabilities is confronted with a number of structural and operational barriers, including resource allocation and the management of incentives for academics (Caldera-Serrano, 2019).

In Panama, public universities have established various policies and mechanisms to encourage research among their teaching and research staff. These policies range from the prioritization of research in university statutes to the implementation of specific incentives, such as reduced teaching loads and funding for research projects. These initiatives are imperative in overcoming the limitations imposed by the absence of a robust state policy, lack of government interest and the insufficient allocation of state funds dedicated to research.

However, despite institutional efforts, there are persistent barriers that limit the effectiveness of these existing policies. These barriers include internal cultural issues, the need to improve organizational structure, competition for resources, research infrastructure, and intellectual property management, among others. A thorough consideration of these factors is essential to comprehensively understand the research landscape at Panama's public universities and formulate strategies that can effectively strengthen the capacity and quality of scientific research in the country.

1.1 Background

Panama's public university system is comprised of five institutions, which differ in size, focus, scope and research practices. Their role in shaping the scientific research ecosystem is pivotal, as they represent the largest share of human resources dedicated to higher education and scientific production in the country. The following table (Table 1) summarizes the main characteristics of these five public universities in Panama:

This section focuses on the general current state of faculty research motivation and barriers to do research in public universities. Policies and culture are as well described. Finally, an overview of actual institutional policies, specifically in Panama, is given.

1.1.1 Faculty motivation, incentives and barriers

In the Panamanian context, academics' motivation to engage in research activities is influenced by specific institutional and regulatory factors (Barsallo et al., 2025). However, despite the existence of government agencies designed to promote academic engagement in research initiatives, empirical studies have revealed that the practical implementation of these incentives faces great challenges. Sánchez-Galán (2020) documented that budgetary constraints, excessive teaching hours, and a lack of adequate infrastructure are the main barriers limiting faculty participation in research activities, highlighting a gap between regulatory aspirations and institutional reality. Subsequently, Barsallo et al. (2025) concurred that work-related issues, including class schedules, combined with bureaucracy, salaries, lack of opportunities, and job stability, have been identified as the primary demotivating factors for academics in research.

From another perspective, Batista (2025) asserts that despite recent technological innovations and new teaching/learning methods,

TABLE 1 Key characteristics of Panama's public universities.

University	Institutional focus	Enrollment (2023)	Faculty	Scientific production/ rankings/strengths	Territorial coverage
Universidad de Panama (UP)	Comprehensive public university with broad professional and technical programs in humanities, sciences and social fields	89,783	4,731	• Highest national scientific output, • ~ 41 registered researchers	Nationwide presence; multiple campuses; largest enrollment
Universidad Tecnologica de Panama (UTP)	Specialized in engineering, technology and applied sciences	25,763	1,892	• Strong in innovation, and research visibility • ~ 36 registered researchers	Multiple campuses across major regions
Universidad Autonoma de Chiriqui (UNACHI)	Regional institution located in Chiriquí with diverse academic programs	22,185	1,076	• Presence in national assessments of scientific production and impact • ~ 9 registered researchers	Main campus in Chiriquí; serves western Panama
Universidad Especializada de Las Americas (UDELAS)	Specialized in health, rehabilitation, applied sciences and education	12,294	1,306	• More modest scientific output compared to UP and UTP • ~ 2 registered researchers	Main campus in Panama City and specialized extensions in western Panama
Universidad Maritima Internacional de Panama (UMIP)	Specialized in maritime sciences, nautical studies, naval engineering and marine transport	1,491	60	• Small number of researchers; specialized scientific focus • ~ 3 registered researchers	Focused presence in strategic maritime sector

Enrollment and faculty data correspond to 2023 estimates; coverage refers to institutional presence within the national territory. Own elaboration based on National Institute of Statistics and Census (Instituto Nacional de Estadística y Censo, 2025) publications on education, year 2023 and descriptions found in Reisberg (2021).

TABLE 2 R&D expenditure, funding structure and personnel in Panama (2014–2023).

Count	Indicator	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
	R&D Expenditure (million USD)	109.84	107.97	148.76	169.17	180.70	191.63	269.65	237.79	291.93	280.45
	R&D Expenditure (% of GDP)	0.12	0.11	0.13	0.14	0.14	0.14	0.25	0.18	0.18	0.18
	Researchers (headcount)	451	525	537	616	581	623	1,210	1,127	780	1,264
Funding origin (% of total)	Government	35.9	30.8	47.7	46.8	44.1	41.9	48.4	48.1	53.7	53.0
	Enterprises (public and private)	1.2	1.1	0.7	1.1	1.6	2.0	1.0	1.0	1.6	0.9
	Higher education institutions	0.1	0.7	9.2	13.9	17.7	15.9	21.2	21.2	19.5	19.9
	Non-profit organizations	4.2	4.0	0.1	1.1	1.1	1.8	5.0	7.9	5.3	5.2
	foreign	58.6	63.4	42.3	37.2	35.6	38.4	24.4	21.9	19.9	21.1

Source: OEI and UNESCO (2025). El Estado de la Ciencia 2025, pp. 106–123.

major contradictions remain, such as the lack of appropriate infrastructure and an education system that does not sufficiently stimulate critical thinking or align with the demands of the local labor market.

1.1.2 Institutional policies and research culture

Panama aims to strengthen its research culture by engaging students in research activities early in their university education. Panamanian legislation has had a positive impact on the academic landscape by recognizing teaching, research, outreach, and administration as key indicators of quality in higher education. However, the institutional focus remains predominantly on teaching rather than research, which limits the impetus to develop robust research (León, 2020).

In an effort to promote a culture of research in the country, various initiatives have been developed. For instance, the Panama Declaration on Open Science establishes guidelines that promote open access for individuals with the capacity to obtain and disseminate educational materials, data, and methods, among other objectives. This initiative aims to foster collaboration between universities and civil society actors, recognizing the impact of applied science on addressing local problems. This initiative signifies a cultural transformation toward a more inclusive and transparent research practice (Tejedor De León, 2021).

Another initiative is the Scientific Initiation Day (JIC), as a first formal experience for students to be engaged in research. This was based on the principle that universities are not merely degree-granting institutions but vital centers for science, technology, and innovation. The program, ongoing since 2015, seeks to holistically develop students, professors, researchers, and administrators by fostering research, knowledge production, and critical thinking. Aunado a esto, también están los congresos de las universidades, y las aproximadamente 38 revistas científicas pertenecientes a universidades panameñas que operan bajo el modelo acceso abierto diamante que permite la publicación de investigación científica sin costos y el libre acceso a los lectores de acuerdo con PANINDEX.

While these institutional policies and cultural initiatives represent important steps forward, their actual impact on Panama's broader research landscape requires closer examination.

1.1.3 Panama's research landscape

According to Nevache (2019), Panama's quality of research institutions qualifies it with a minimal remark on the global competitiveness index. To this, it must be said that Panama's research landscape in public universities today has positively been shaped by both national initiatives and internal institutional dynamics, with varying levels of success in promoting research activities and fostering a research culture. The National Secretariat of Science and Technology (SENACYT) has been a pivotal player in the country's research development, providing funding and facilitating collaboration between universities and international entities. According to a study by Soto, by 2020 Panama had registered 37 academic journals in digital format; 33 of these journals are university journals (Soto, 2020). The study emphasizes the creation of bioethics committees and efforts to get better international visibility, which are critical to strengthening Panama's research capacity. However, the lack of a cohesive national policy and the fragmented implementation of initiatives hinder further progress (Soto, 2020; Esquivel, 2019; Svenson et al., 2018).

Furthermore, national indicators (Table 2) highlight the structural limitations of the country's research system: Panama invests only 0.18% of its GDP in R&D, one of the lowest rates in the region, while counting with just 0.47 researchers per 1,000 members of the economically active population. Although there has been incremental growth in R&D expenditure over the last decade, funding still depends largely on government and international sources, which hinders the sustainability of scientific research development (Organización de Estados Iberoamericanos para la Educación, la Ciencia y la Cultura and UNESCO, 2025).

Although R&D expenditure has increased, reliance on government and foreign financing, together with minimal private-sector investment, raises concerns about the long-term sustainability and autonomy of national scientific development. Similarly, fluctuations in the number of researchers over time suggest a fragile human capital structure that is highly dependent on external conditions rather than a consolidated national research capacity. Hence, understanding how university researchers experience these conditions becomes a necessary step for this study.

2 Materials and methods

In order to address these complex efforts needed and understand the complicated nature of research development in Panama's public universities and its actors perception, this study uses the qualitative approach, which is following explained.

This qualitative, exploratory, and cross-sectional study adopted an interpretive phenomenological approach (Creswell and Poth, 2018) to

interpret the multiple realities that influence the development of scientific research in Panamanian public universities. Non-numerical methods were used to prioritize meanings and perceptions over quantitative analysis (Merriam and Tisdell, 2016). The methodological design is presented through the constructivist-interpretive paradigm, recognizing that respondents construct diverse meanings about their own research experiences (Lincoln and Guba, 1985).

This study included professors and researchers from four of Panama's five public universities.[1] All participants were selected through purposive and convenience sampling (Patton, 2015), prioritizing those with direct experience in university research and their participation was entirely voluntary. Data collection included the ENACT[2] (Mendoza, et al., 2025) national survey administered to public university faculty (n = 921), semi-structured interviews with university authorities (n = 6), and faculty perceptions gathered in a seminar-workshop (n = 117). Although the ENACT survey involved 921 participants nationally, the present study analyzes responses to a single question completed by 565 respondents. This single-question analysis, combined with interviews and the workshop poll, constitutes the primary data collection for this article. The reference to the full ENACT sample is provided for contextual and methodological transparency.

Although all participants are affiliated with Panamanian public universities, the study focuses on the perceptions, experiences, and arguments of professors and researchers as individual actors in the research ecosystem, rather than on an institutional comparison between universities.

To strengthen the credibility and depth of the analysis, multiple triangulation was applied (Denzin and Lincoln, 2018), combining methodological triangulation through a survey, interviews, and an exploratory post-seminar perception poll (Fusch et al., 2018) with data triangulation by approaching various groups of participants (Schlunegger et al., 2024; Carter et al., 2014; Santa Cruz Terán, et al., 2022). Researcher triangulation was achieved through the participation of three independent coders who analyzed the data separately before reaching consensus (Flick, 2018). Data analysis was performed using inductive thematic coding following the six phases of Braun and Clarke (2006): familiarization, initial coding, theme search, theme review, definition and naming, and final report preparation. Atlas.ti software was used for the systematic organization and coding of qualitative data. As an additional strategy to re-inforce methodological rigor, researcher reflexivity was practiced through the use of logs (Finlay, 2002). Criteria of credibility, transferability, dependability, and confirmability (Lincoln and Guba, 1985) were used to ensure methodological rigor. Ethical considerations included approval by the Bioethics Committee, the use of informed consent from all participants, data confidentiality, and their anonymity in the presentation of results.

This research is part of a broader project entitled "Factors influencing academic and scientific research in Panama: From the perspective of professors and researchers" [Factores que influyen en la investigación académica y científica en Panamá: Desde la perspectiva de los docentes e investigadores], of which this article is the fourth output.

2.1 The survey

The ENACT survey (Mendoza et al., 2025) was created and conducted virtually between May and August 2023 as part of the project "Factors Influencing Academic and Scientific Research in Panama: From the Perspective of Professors and Researchers." It involved the participation of 921 academics and researchers affiliated with public universities in Panama. The ENACT questionnaire, with a total of 39 questions, provided key information for the development of several articles; however, for this article, the responses to a single question were analyzed: What is your opinion on the role of re-search in Panama with regard to the sustainability of higher education?. For this question, 565 responses were analyzed.

2.2 The interviews

For the semi-structured interviews, the population was segmented into vice-rectors for research, both current and former, from the participating public universities. The authorities selected for the interviews were identified based on their important role in the formulation and implementation of research policies within their respective universities. The interviews were conducted in person, allowing for direct and detailed interaction, which facilitated the collection of rich and in-depth data on specific experiences and perceptions related to research.

The interviews were conducted using a semi-structured interview guide, which included five key questions designed to explore various facets of the research environment at universities (Kvale and Brinkmann, 2015). The interview questions were designed to elicit responses regarding institutional policies, research incentives, faculty participation in research projects, perceptions of international journal indexing, and other critical factors influencing research.

Each question was designed to inquire into the following specific topics:

- University Policies: Identification and classification of policies that contribute to re-search development, such as funding and inter-institutional collaboration.
- Incentives for Faculty and Researchers: Evaluation of the types of incentives provided and their effectiveness from the perspective of vice-rectors.
- Academics Participation in Research: Analysis of strategies employed to foster active faculty participation in research.
- International Journal Indexations: Assessment of how these indexations are perceived in terms of their impact on research quality.
- Factors Influencing Faculty and Researchers: Identification and categorization of specific factors that affect researchers during the research process.

2.3 The poll

Following the ENACT survey and authorities interviews, and as part of the same project, a complimentary seminar-workshop (SIFESS) was organized to raise awareness among professors about the importance and significance of conducting scientific research. At the

1 Universidad de Panamá (UP), Universidad Tecnológica de Panamá (UTP), Universidad Especializada de las Américas (UDELAS), Universidad Autónoma de Chiriquí (UNACHI), Universidad Maritima Internacional de Panama (UMIP). From these, the last university did not participate in the study.

2 Survey on Academic and Scientific Research (Encuesta Sobre Investigación Académica y Científica, ENACT).

conclusion of the SIFESS seminar, participants were invited to complete a voluntary post-seminar perception poll aimed at capturing participants' immediate reflections following the activity's collective discussion. The perceptions of 117 professors were polled regarding the role of research as a corner-stone for the sustainability of higher education. The poll consisted of open-ended reflective questions addressing the relevance of the seminar content to the university and national context, sustainable development, digital transformation, and the relationship between science policy and academic practice.

The poll was designed for exploratory and contextual purposes rather than formal measurement. It was not subjected to psychometric validation procedures appropriate for quantitative instruments and its methodological rigor was ensured through reflexive question design informed by preliminary data analysis and the administration in a research-designed context (the SIFESS workshop). Responses were reviewed thematically and used to complement the ENACT findings and interview data within the study's triangulation strategy.

It must be said that the poll was not designed as a standardized or validated measurement instrument, nor intended for statistical generalization. Its role in the study was contextual and illustrative. These approaches align with established standards for qualitative research instruments (Merriam and Tisdell, 2016; Patton, 2015).

2.4 Data analysis

The coding process was carried out through an inductive thematic analysis. First, open coding of the testimonies was carried out, identifying units of meaning from a line-by-line reading. Subsequently, the initial codes were compared and grouped into broader thematic categories through an iterative process of comparison and refinement.

For the purpose of data analysis, a qualitative approach was adopted, using content analysis to facilitate a detailed interpretation of the responses obtained from surveys, interviews, and other data sources. Thematic content analysis has been identified as a critical component of a qualitative research approach, distinguished by its empirical-inductive reasoning interpretations of the information gathered through different means in the context of the research (Castro et al., 2022; Moraima Campos and Auxiliadora Mújica, 2008). In this way, content analysis reveals the meaning that emerges from the text, manifested as clear and hidden information. Explicit information is that which a writer expresses directly, while hidden information is tacit and acquires meaning in a specific context. This text analysis methodology, based on processes of decomposition and classification of implicit information that acquires meaning in a given context, facilitates the identification of different types of documents, such as interview reports, observation protocols, field notes, photographs, television advertising, newspaper and magazine articles, political speeches, among others. In this type of study, fundamental and essential aspects such as interpretation and inference were highlighted (Díaz Herrera, 2018).

Data analysis followed a thematic coding process. In this process, initial codes were generated through the close reading of transcripts, and identifying recurrent meanings and concepts. These codes were subsequently grouped into broader thematic categories through an iterative process of comparison and refinement.

Next, the responses were coded and categorized using the Atlas.ti data analysis tool, which facilitated the identification of common themes and differences between participants' perspectives. To ensure methodological rigor, credibility criteria were ap-plied through triangulation of sources and validation with key interviewees (Lincoln and Guba, 1985). Reliability was ensured through the use of detailed documentation of the analysis process and systematic coding in Atlas.ti (Korstjens and Moser, 2017). Further-more, a reflective journal was kept to document methodological decisions and acknowledge the researcher's influence on data interpretation. This approach also enabled the exploration of the relationship between institutional policies and their actual impact on researchers' behavior and motivation.

Data and source triangulation techniques were used for this study, as various sources of information were used and data collection was carried out using different tools. Triangulation is a research approach that integrates various data sources, theories, and research methodologies to ensure the better understanding and accuracy of a study's data, analysis, and conclusions (Forni and De Grande, 2019; Feria Avila et al., 2019).

The combination of detailed interviews, survey data, and a rigorous data analysis process using the Atlas. Ti program ensured that the study's findings provides an integrated view of the factors influencing research at Panama's public universities.

For the purposes of translation, refining writing and text editing, generative artificial intelligence (GenAI) has been used in this paper.

3 Results

The results presented below derive from the thematic analysis of the participants' statements, in which recurring patterns, discursive tensions, and divergent perspectives were identified, based on the frequency of arguments, the intensity of values, and the recurrence of concepts within the collected testimonies.

3.1 Interviews to university authorities

The analysis of interviews conducted with authorities from various Panamanian state universities was organized according to the categories of questions asked. The analysis was conducted with the anonymity of institutional staff in mind. The categories of questions asked included institutional policies, incentives offered, faculty participation in research, evaluation of international indexing, and factors that influence researchers and professors. The following findings derived from recurrent patterns identified across participants' narratives, reflecting shared perceptions.

3.1.1 Institutional policies

Participating universities formally recognize research as a primary activity, prioritizing the allocation of resources over other activities. Concurrently, the development of research, the establishment of research groups, and policies designed to not only support these initiatives but also ensure their sustainability are being promoted. Consequently, the establishment of research groups is being promoted by authorities nationwide through regional centers, extensions, and various interdisciplinary, interinstitutional, regional, and international groups.

However, it is necessary to point out the lack of state support for research in the country, as well as the need to develop a state policy

for research at the national level. Despite the existence of a national entity that performs these functions, the role and effectiveness of this entity in this area require further examination. This issue constitutes a priority barrier for research in the country.

Public universities prioritize research as a mechanism for the country's development, evidenced by their internal efforts to strengthen their regulations and standards. These measures are designed to encourage scientific research activities among teaching staff.

3.1.2 Incentives for research

While there is a consensus on the importance of research development and the need for strengthening it within universities, is not sufficient. Having regulations is a step, but it is still insufficient even knowing that resources tend to be inadequate. Therefore, a variety of incentives are in place to motivate professors to engage in re-search, establish networks and/or research groups, and respond to public calls from the National Secretariat of Science, Technology and Innovation (SENACYT), the preeminent national entity in this domain. Among the incentives mentioned by the inter-viewed authorities were teaching load allocations for research (faculty teaching load reduction), the creation of positions such as "special professor with doctorate" to recognize and support the dedication to research of those professionals who have obtained their doctoral and/or postdoctoral education degrees and enter universities to work as faculty. Likewise, some universities have opted to create within their organizational structure the position of researcher and/or emeritus researcher.

Other mentioned incentives refer to promoting support through the allocation of funds to support researchers' participation in international events. This type of activity should be regarded as an investment to stimulate the management of researcher networks and the strengthening of research projects that respond to the country's needs.

In this regard, university authorities made evident that they are going through a paradigm shift in the role of universities through a change in institutional culture with respect to research. This shift in perception is indicative of a change in the perceived importance of research, which has been fostered by the implementation of new incentive policies.

3.1.3 Academics participation in research

Regarding faculty participation in research, from the perspective of university authorities, the implementation of incentives and the momentum being given to the development of scientific research in universities has enabled acceptance by professors through the creation of research groups. An example of this is the creation and recognition of a growing number of research groups in recent years, allowing for more autonomous and efficient management of the research being conducted. As for participation in events and publications, faculty have achieved financial support for presenting papers at international conferences, encouraging visibility and active participation in the global scientific community, which in turn impacts the creation of networks, a greater number of publications in high-impact journals, and visibility not only for the universities but also for the researcher-professors and their contribution to the country.

Regarding evaluations local scientific journals with international indexations, the opposition of opinions existing among academics remains open for reflection; that is, the ongoing discussion about the commercial model versus open science, with a focus toward independence in knowledge management and its recognition. Participants' views reveal divergent interpretative positions toward the value of local scientific journals vs. international indexation, indicating there are persistent contending views between institutional recognition mechanisms and epistemic legitimacy. However, the presence of contradictory evaluations should not be viewed as a flaw in the study, but rather as a discovery that highlights the disparity of criteria among the academic community regarding models of scientific validation.

Another aspect that emerges in this area relates to the importance of participating in Scopus database management workshops. Likewise, other training programs could be included that strengthen the researcher role, their research processes, article development, and citations.

3.1.4 Factors influencing research

The perception of university authorities (Table 3) regarding the factors that influence research is related to institutional evolution and paradigm shifts, since models have had to be adapted to incorporate research as an integral part of the university's mission, seeking a balance between commercial and open science paradigms. As Paradigms shifts involve a process and incentives until the community not only accepts them but also incorporates them, participates in them, and in turn becomes a pillar that motivates others to collaborate in research, one of the reasons why research is limited could be related to project funding and/or participation in international conferences and, it is in this context, that research policies being implemented are directed.

In this sense, cultural barriers and resistance to change are present among the teaching staff. Therefore, it is necessary to confront, motivate, and challenge this culture from within the university, and to implement structural changes to foster an environment conducive to research. This task represents a critical component of the re-search policies that must be promoted to stimulate the development of research. It is also expected that faculty members will embrace these policies as part of their university activities.

What has been observed in public universities is the notable existence of similarities in their strategies to promote research. These institutions have implemented incentive policies that include reduced teaching hours, allowing faculty members who conduct research to distribute their time in such a way that they reduce their teaching hours in order to devote more time to their respective research projects. Furthermore, these universities have exhibited a commitment to the expansion, strengthening and consolidation of their research infrastructures through the creation and support of re-search groups, as well as by promoting participation in scientific events at both the national and international levels. This integrated approach underscores a shared vision toward strengthening research capacity and knowledge generation, demonstrating a concerted effort to overcome structural and cultural barriers and improve the quality and impact of research in the Panamanian academic landscape.

3.2 The ENACT survey

Respondents in the ENACT survey commented on the relevance and prominence of research in the sustainability of higher education (Figure 1).

TABLE 3 Summary of the results of interviews with university authorities.

Category	Emergent themes
Institutional policies	• Legislative priority for research. • Absence of robust and effective government policies. • Reactivation of research programs. • Regulations for research groups. • Extension strategies to regional centers.
Research incentives	• Course load reductions and research funding. • Creation of new academic positions (researcher positions). • Promotion of practical applications and intellectual property. • Support office for project management and financial support for international conferences.
Faculty participation	• Research seedbeds for students. • Recognition for outstanding students. • Support for events and publications. • Integration of teaching and research.
Indexation evaluation	• Adoption of open science and internal initiatives. • Aspirations to achieve indexations. • Exploration of funding alternatives. • Recognition through academic awards.
Influencing factors	• Influence of commercial and open paradigms. • Participation in international projects. • Challenges in workload distribution. • Continuous improvements in policies. • Importance of optimal conditions for research.

In the content analysis of participants' responses, seven major themes were identified. Here are some examples of respondents' observations and experiences.

3.2.1 A positive role

> "The role of research is a variable linked to the budgetary capacity assigned by the executive body; if the budget is limited, the role is also limited." (Male professor, Medicine, 66 years old, translated by the author).

> "We must continue making the necessary efforts to change this for the benefit of all and to achieve greater independence from the State." (Female professor, Languages, 50 years old, translated by the author)

3.2.2 A work in progress

> "It is improving compared to the work carried out by previous administrations, but we must continue working to guarantee the appropriate conditions and pursue academic excellence—from infrastructure to the availability of research funds." (Male professor, Exact Sciences, 35 years old, translated by the author)

3.2.3 Research requires more implementation, dissemination, and promotion

> "It would be beneficial for research to ensure its social relevance. To this end, it should be managed in a way that fits the needs of the society in which HEIs are located. In addition, the results could be used to improve the academic offer." (Female professor, Medicine, 42 years old, translated by the author).

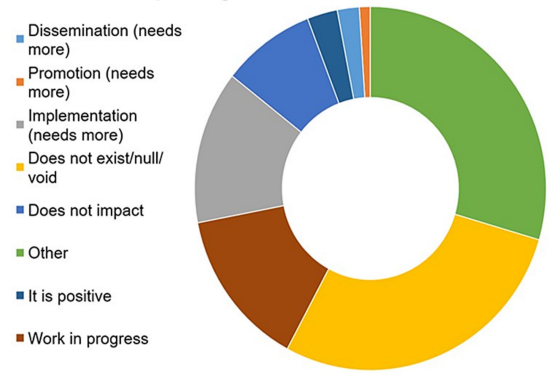


FIGURE 1
Perception of the protagonist role of research in the sustainability of higher education.

> "As individuals interested in scientific progress, it is important to keep ourselves informed about advances in our field of interest, participate in science outreach activities, and actively advocate for the value of science in society." (Female professor, Education, 43 years old, translated by the author).

> "The promotion of research in higher education can play a crucial role in the sustainability of the country, as it can contribute to solving national problems, improving both basic and higher education, achieving equitable economic development, and fostering sustainable practices, while also allowing students to enjoy doing research rather than simply fulfilling a course requirement." (Female professor, Natural Sciences, 46 years old, translated by the author)

3.2.4 An absent or null role

"There is no significant role. It remains a major issue, as the percentage of researchers must increase substantially to train more research students. The local reality does not benefit from research results due to their low impact, that is, the weak university–community relationship in research. We perform well in international indices, but we struggle with national impact." (Male professor, Economics, 58 years old, translated by the author)

3.2.5 No impact

> "If research played a leading role in decision-making on public policies in health, justice, and education, we would live in a different reality. Panamanian researchers need to transcend the limitations of popular culture and the deficiencies of our educational system in order to succeed in higher education programs abroad or in local programs with certain links to foreign educational or scientific institutions. In Panama, research does not play a leading role." (Male professor, Natural Sciences, 42 years old, translated by the author).

3.3 SIFESS seminar-workshop

This event addressed the discussion concerning the actions required to promote and increase the role of research in the sustainability of higher education, a topic of utmost importance that requires careful evaluation and meticulous planning.

The categories represented in Figure 2 correspond to the discursive frequency and thematic centrality identified in the analyzed testimonies. Participants shared their opinions and perceptions, which included themes such as.

3.3.1 Research culture

Respondents highlighted the need to adopt a research culture as an essential part of the new era in higher education. Some of their comments were:

> "A culture of innovation and entrepreneurship can be achieved through incentives for the creation of science and technology-based companies and the development of policies that promote the transfer of knowledge from academia to the productive sector." (Female professor, Social Sciences, 48 years old, translated by the author).
>
> "Promoting collaboration, transparency, and ethics in scientific research, supporting practices such as open access publishing and data sharing, which contribute to the advancement of knowledge and trust in science." (Male professor, Engineering, 32 years old, translated by the author)

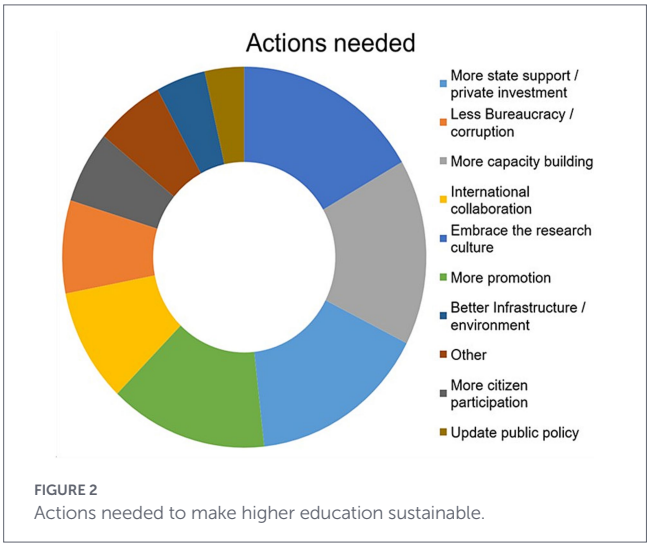


FIGURE 2
Actions needed to make higher education sustainable.

3.3.2 Infrastructure

Respondents evaluated the importance of infrastructure innovation for the proper functioning of the institution's organization. This includes services, technical resources, and facilities that enable the development of research. One interesting comment on this topic was:

> "Achieving better communication with government institutions and private companies to support them with material resources, infrastructure, and human resources… ensuring the conditions and resources for professors to conduct research; therefore, an important aspect is time, otherwise it will be very difficult to do so." (Female professor, Social Sciences, 69 years old, translated by the author)

3.3.3 Scientific literacy

Inferences regarding the origin of the knowledge acquired and transmitted. Such findings can be associated with the trend of decolonization of knowledge.

> "Teaching through practice and not only with theory that comes from other places, but with the theory that we ourselves discover and implement for our well-being." (Male professor, Engineering, 57 years old, translated by the author)

3.3.4 Bureaucracy

A great amount of comments referred to the continuous bureaucracy that can be found in public institutions. However, this simple one encompasses the idea.

> "We must create structures and institutions that easily enable the development of research, without administrative and bureaucratic constraints." (Informant, translated by the author)

3.3.5 International collaboration

> It is acknowledged the need of multidisciplinary cooperation and coordination between academics, business, and the government.
>
> "Collaboration between different scientific disciplines and between academia, industry, and government to address complex problems from multiple perspectives and generate innovative solutions." (Male professor, Engineering, 56 years old, translated by the author)

3.3.6 Citizen participation

The low social valuation of science and the lack of active commitment of the State and the private sector to research.

> "As citizens, we must understand the importance of science in our daily lives. Participate in talks, workshops, and scientific events to raise awareness among the public and ensure that the government and private companies collaborate by investing in research and development." (Female professor, Fine Arts, 64 years old, translated by the author).

These evidences reveal a complex landscape of advances and persistent challenges that merit deeper analysis in terms of organizational and institutional dynamics.

4 Discussion

The discussion on research policies and practices at public universities reveals a series of similarities and differences that highlight both the challenges and advantages in the academic research landscape in Panama.

4.1 Research culture: institutional advances and resistance

Interviewed authorities highlighted the incorporation of research as a priority function in their statutes, the drive toward the formation of research groups, and the promotion of open science. Examples of such policies include those that facilitate the creation of interdisciplinary teams and their extension to regional centers. However, academics have expressed concerns that the research culture has not yet been fully integrated into the institution's core functions. As one respondent indicated in the post-seminar poll, "Research is not regarded as a component of university culture; it is regarded as an additional assignment, not as an integral element of university education" (translated by author).

García-Arce et al. (2021) observed similar disparities on the culture of research in Latin America, which requires going beyond regulatory changes; rather, it demands symbolic and organizational transformations that acknowledge academics as active producers of knowledge. Detach between current policies acknowledgement and actual practices suggests that cultural assumptions about research priorities require systematic attention beyond regulatory changes.

In Villadiego Sánchez (2020) view, the dynamics of organizational culture are contingent upon the exigencies of specific contexts, the strategic initiatives employed for the cultivation of members' capacity for adaptation, and the ability to address the issues posed by advancements in communication technology and the generation of scientific knowledge through research. However, it should be noted that, despite of being the present era marked by important advancements in science and technology, hierarchical structures persist within academic institutions, posing obstacles to the acknowledgement and therefore integration of research into the academic landscape. Even so, Pereira-Medina et al. (2016) have indicated that in order to achieve changes in the organizational culture of higher education institutions, it is necessary to go beyond simple regulatory reforms and undertake transformations that adopt new perspectives on the reality and daily life of academics' work.

4.2 Academic incentives: between the norm and practice

Along with the cultural aspects, the implementation of incentives presents another layer of complexity. Results from the present study suggest that public universities have implemented incentives such as reduced teaching hours, funding for conference attendance, and the creation of specialized academic positions. Nevertheless, concerns regarding the coverage and sustainability of these actions persist: "The incentives exist, but they do not reach everyone. Sometimes they seem more like rewards than systematic policies" (university authority interviewed).

According to the SDT (Self-Determination Theory) model proposed by Ryan and Deci (2020), external incentives received by academics, such as reduced teaching hours or funding, if perceived as controlling rather than supportive, can undermine intrinsic academic motivation by frustrating the needs for autonomy and competence.

According to García de Fanelli (2019), the formulas for distributing budgetary resources are based on input and output indicators in order to promote reforms in universities; however, there are still efficiency issues related to the coverage and sustainability of funding for university development.

The regional outlook corroborates these limitations. Data from United Nations Economic Commission for Latin America and the Caribbean (ECLAC) (2023) states that, "while the United States, Japan, and the Republic of Korea invest between 2.5 and 3 points of their gross domestic product (GDP) in research and development, and the European Union invests almost 2 points, our continent as a whole invests only 0.5 points of GDP." Organizational Culture. This perception supports the notion that, within the Latin American context, incentives for research lack sustainable financing, depend on short-term decisions and are susceptible to the political currents of the moment. In line with this, research shows that external rewards can erode motivation when they impede autonomy, competence, and relatedness. When academic staff perceive incentives to be short-term or inconsistent, the resultant effect may be the decrease of intrinsic motivation and a reduction in sustained engagement (Szulawski et al., 2021).

4.3 Social impact and research visibility

The ENACT survey (Mendoza et al., 2025) revealed that several respondents believed that academic output has a greater impact on international rankings than on solving national problems. As one ENACT respondent expressed, "We perform well in international rankings, but have little real impact on public policy or communities". This could be attributed to the high performance standards demanded for the publication of scientific articles in prominent indexed journals, a strategy promoted to raise the quality of university output and thus improve the institution's reputation, as well as contributing to the academic development of its researchers.

This strategy, regarded as unrelated to local reality, is not necessarily negative. However, it would be worthwhile to consider the scope that this scientific production achieves when its focus is not only on revealing new knowledge, but knowledge that has had an impact in the place where it was produced. In this regard, Coronel (2023) and

Bayas Romero et al. (2022) posit that, in order to be socially relevant, universities must focus primarily on understanding the difficulties encountered in the processes of globalization and taking advantage of its opportunities, maintaining a forward-looking perspective that reaffirms their role in research and their commitment to generating tangible benefits for society through their educational processes and social development.

The interplay between global visibility and local relevance reflects present-day challenges in research impact assessment. The concept of 'social impact' has evolved beyond citation metrics to include broader social, economic, and policy impacts (Greenhalgh et al., 2021). This reinforces an explanation for why research that performs well in international rankings may still be perceived as disconnected from national development agenda. On this regard, Pascual Del Rosario (2024) adds that while Panama exhibits progress in higher education access indicators, the impact on the Social Progress Index (Krylova et al., 2025) remains limited. This indicates that academic outcomes such as graduation rates, publications or recognitions do not consistently produce sustainable improvements in social welfare. As it is, this gap reinforces the need to articulate university research with public policies and projects that respond to the country's actual demands. It is this same belief that universities must strengthen the social relevance of research, articulating scientific production with public policies and local development, which highlights the necessity of reinforcing "science with social consciousness," integrating the communication of results to non-academic audiences.

4.4 Structural barriers and state weaknesses

Structural barriers at the national level further aggravate the difficulties associated with visibility. Interviews with university authorities revealed persistent obstacles such as faculty overload (class hours), inadequate infrastructure, and administrative bureaucracy. In addition to this, the absence of a robust and coherent national policy that supports university research as a cross-cutting factor, as one interviewee said: "Having regulations is a step, but it is insufficient… the lack of resources and bureaucracy limit their implementation" (University authority).

At the macroeconomic level, state investment in science and technology is note-worthy. In 2024, it represented only 0.18% of GDP, a figure that falls below the regional average, reflecting a structural limitation in public funding for research (Senacyt, 2024). This phenomenon has been documented in Latin American universities, which are confronted with constraints that impede their capacity to serve as effective conduits for the dissemination of knowledge that can inform public policy. These limitations stem from systemic barriers that impede the effective transfer of research findings (Palacios, 2023).

Miller et al. (2023) argue that, as in other countries in the region, limited state investment in science and technology combined with administrative centralization restricts institutional research capabilities in Panama. Palacios (2023) confirms this, adding that other aspects are also important to consider, such as institutional politicization, limited efficiency and effectiveness in communicating with decision-makers, poor media relations, differences in the skills of new researchers, government entities' well-known bureaucracy, social outreach as a strategic focus in the institution's mission, and the lack of interdisciplinarity.

All these structural constraints are consistent with the principles of Resource Dependency Theory as applied to higher education institutions, as the capacity for research is linked to the resources available. This scenario can produce resources dependence that can hinder institutional autonomy and the establishment of research priorities (Pfeffer and Salancik, 2015). The limited state investment in research and development causes resource scarcity that forces universities to compete for limited funds, which can always affect research quality and sustainability.

4.5 Tensions among stakeholders: authorities vs. academics

Disagreements between university authorities and academics often stem from differing views on autonomy, resource allocation, and the establishment of research priorities, which can result in the ineffective implementation of institutional policies.

The application of data triangulation techniques resulted in the identification of major discrepancies between the perceptions held by authorities and those held by academics. While authorities emphasize advances in regulations and structures, academics express skepticism, saying "We don't know if these changes are sustainable or just part of a passing fad" (SIFESS post-seminar respondent).

The tense situation arising from the resources allocated to the institution and how it manages to get them distributed to the research area is not unique to Panama, but rather is a phenomenon that extends to Latin America as a whole, according to Villalba Benítez (2017). This phenomenon is exemplified by government budget cuts, which often result in the university and its products being relegated to a secondary status in the event of a local crisis that needs the limitation of state resources. From the perspective of Institutional Theory, these measures reflect not only material constraints but also the way universities adapt to broader socio-political expectations. Institutions often comply with governmental pressures and resource limitations in order to maintain their legitimacy, even if this compliance undermines their academic mission (Meyer and Rowan, 1977).

According to Araujo (2020), the tensions that arise between teaching, research, and academic outreach highlight the differences in priorities and expectations among institutional actors, giving rise to conflicts that are managed according to the cultural practices and traditions of each university. There are also those who posited that, in the absence of faculty involvement in the formulation and execution of research policies, these policies may be perceived as external and temporary impositions, rather than as collaborative strategies.

4.6 Implications for university and national policy

The data show that strengthening research in Panamanian public universities requires a systemic approach that integrates institutional, state, and social dimensions. This aligns with what Valdés-Montecinos and Ganga-Contreras (2021) argue stating that university governance requires coordination between internal and external actors, reinforcing institutional legitimacy and promoting participatory and transparent management. This proposes a sustained transition toward participatory university governance models, characterized by the consistent allocation of resources and the formulation of long-term strategies. Dependence on ineffective state policies and the lack of specific funding for research are common obstacles that hinder researchers' ability to conduct impactful research. Moreover, the constrain of

effectively integrating research and teaching remains an area that requires attention, as balancing these two functions can be complex given administrative and academic demands.

Strengthening research in Panamanian public universities requires adopting contemporary Shared Governance Models that move beyond traditional hierarchical structures. Recent developments in higher education governance emphasize collaborative approaches to decision-making that involve multiple stakeholders in meaningful ways, rather than soliciting input after decisions have already been made (Kezar and Holcombe, 2017). This approach is relevant to the current situation in Panama given lack of connection between authorities and faculty identified in this current work.

5 Conclusion

Given the qualitative nature of the study, the presented findings do not seek statistical generalization, but rather provide analytically grounded insights into predominant participant observations and experiences within the public university research ecosystem. This qualitative study examined the complex landscape of research policies, incentives, and barriers in Panama's public universities through multiple theoretical lenses in the discussion. The triangulated analysis revealed that while important policy advances have been implemented, there is a lack of connection that persists between institutional aspirations and academic realities, requiring systemic transformation rather than incremental reforms. These advances are reflected in concrete actions reported by authorities, such as the creation of formal research units, reductions in teaching load, and financial support for international conference participation.

Our analysis had several limitations. Since the study only focuses on public universities, it may not reflect the private sector dynamics. The use of qualitative approach, while providing deep insights, limits generalizability provided by the quantitative approach in a mixed method study. Future research should examine private university experiences, consider longitudinal impact assessments, and explore cross-national comparisons within the Latin-American context.

Results indicate that academics and researchers believe adjustments should be made by strengthening the current academic structure, cultivating a culture of research, and creating spaces to disseminate results. This would allow scientific communication to reach society in an understandable manner and contribute to decision-making. The data also show that academics and researchers consider that adjustments should be made by strengthening the current academic structure, cultivating research as a culture, and generating spaces for the dissemination of results so that scientific communication reaches society in an understandable manner and contributes to decision-making on higher education development, making it more sustainable. The reduced level of engagement observed in the original sample and subsequent responses to this question may be due to indifference toward the subject matter or not knowing what to say.

To overcome some of these barriers, it is suggested that these institutions could benefit from greater collaboration with the government and other entities to develop more effective policies that financially support research. In addition, the implementation of more rigorous audit systems could help ensure the efficient use of resources dedicated to research. Likewise, more tangible and diversified incentives could motivate professors to participate more actively in research projects. There is also a positive confirmation that Panamanian public universities have made some important advances in creating policies, incentives, and structures that promote scientific and academic research. The establishment of research groups, the implementation of reduced teaching hours for faculty, participation in national and international conferences, and the incorporation of specialized academic figures are examples of actions aimed at strengthening the research ecosystem. However, these initiatives, while valuable, still face considerable issues in terms of coverage, equity, and long-term sustainability.

The evidence also reveals discrepancies in the perceptions between university authorities and those of professors and researchers. While authorities highlight regulatory advances, the creation of support structures, and improvements in research infrastructure, academics doubt the continuity, scope, and effectiveness of these measures. Even though participants recognize progress in the implementation of research policies and structures, the so-called progress is still perceived unevenly and frequently as insufficient compared to the real demands of academic work. This gap reflects not only differences in perspective but also the need for stronger, more effective channels of dialog and participation. These channels are necessary to develop research policies with a shared vision and sense of shared responsibility.

Although scientific output has achieved greater visibility in international indexes, its impact on solving local problems and formulating public policies remains limited. The imbalance between global projection and local relevance highlights the need to redirect research toward dual purposes: contributing to the global academic community and responding effectively to the country's needs. In this sense, research at Panama's public universities should be consolidated as an engine for socioeconomic and cultural development, collaborating with the government, business, and community sectors.

To achieve these priorities, it is essential to strengthen university governance by encouraging academics to actively participate in designing and evaluating policies. This requires ensuring sufficient and sustained funding through coordinated efforts between the state, universities, and the private sector. Additionally, incentives should be diversified to promote equity and ongoing motivation, while reducing bureaucratic barriers that limit research productivity. Similarly, it is crucial to promote scientific communication by adapting research results to make them accessible and understandable to society, thereby contributing to informed decision-making.

Taking into account the barriers and opportunities presented in the study, several key conclusions and recommendations emerge for strengthening research in Panama's public universities. Although all participating universities were committed to improving research, there are still areas where cooperation and institutional reforms could significantly increase their impact in academic and scientific fields. It is also important to continue disseminating and encouraging research development and the formation of new research groups to support a paradigm shift and research culture in academic communities.

Strengthening research in Panama's public universities cannot be conceived as a series of isolated initiatives, but rather as an integral strategy combining quality, relevance, and social impact in order to encompass deeper institutional change. This study's theoretical and empirical contributions demonstrate that sustainable research development demands coordinated interventions addressing cultural assumptions, structural constraints, and stakeholder relationships

simultaneously. Only through such integral transformation can Panama's public universities fulfill their potential as engines of socio-economic development while contributing meaningfully to global knowledge production. To move forward, the commitment of many stakeholders is needed. This commitment must be based on data-driven policy-making and a solid understanding of how organizations change.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

The studies involving humans were approved by the Ethics Committee of the University of Panama (No. CBUP/170/2022 del 1 de junio de 2022). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

GB: Conceptualization, Data curation, Formal analysis, Funding acquisition, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. EM: Data curation, Software, Validation, Writing – original draft. RC: Conceptualization, Data curation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing. ED: Investigation, Software, Validation, Writing – original draft.

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References

Araujo, S. (2020). *Tensiones entre docencia e investigación universitaria en Argentina. Una mirada retrospectiva*: Universidad Nacional del Centro de la Provincia de Buenos Aires, 63–109. Available online at: https://www.memoria.fahce.unlp.edu.ar/libros/pm.7229/pm.7229.pdf

Barsallo, G., Mendoza, E., Arboleda, M., Torreiro-Casal, M., and Ordoñez, C. (2025). Academics' motivation to research: a study on public universities in Panama. *Front. Educ.* 9:1465824. doi: 10.3389/feduc.2024.1465824

Batista, M. (2025). El impacto de la crisis socioeconómica en la educación universitaria en Panamá: 2020-2024. *Re-vista Contacto* 4, 289–285. doi: 10.48204/contacto.v4n2.6697

Bayas Romero, E. L., Coloma Coloma, L. H., Gaibor Yánez, Í. D., and García García, V. T. (2022). La investigación educativa y su pertinencia social en las Instituciones de Educación Superior. *Pro Sciences: Revista De Producción, Ciencias E Investigación*, 6, 203–210. doi: 10.29018/issn.2588-1000vol6iss42.2022pp203-210

Benavides, M., and Gómez, C. (2005). Métodos en investigación cualitativa: triangulación. *Rev. Colomb. Psiquiatr.* 34, 118–124. Available online at: http://www.scielo.org.co/scielo.php?script=sci_arttext&pid=S0034-74502005000100008&lng=en&tlng=es (Accessed February 18, 2026).

Braun, V., and Clarke, V. (2006). Using thematic analysis in psychology. *Qual. Res. Psychol.* 3, 77–101. doi: 10.1191/1478088706qp063oa

Caldera-Serrano, J. (2019). Análisis de las políticas de I+D+i en Panamá implementadas por SENACYT. *Rev. Iberoam. Cienc. Inf.* 12, 956–975. doi: 10.26512/rici.v12.n3.2019.24441

Carter, N., Bryant-Lukosius, D., DiCenso, A., Blythe, J., and Neville, A. J. (2014). The use of triangulation in qualitative research. *Oncol. Nurs. Forum.* 41, 545–547. doi: 10.1188/14.ONF.545-547

Castro, C., Labra, O., and Chamblás García, I. (2022). El análisis de contenido temático: Una mirada a sus etapas desde Nvivo12©. *Revista Internacional de Ciencias Sociales Interdisciplinares* 10, 143–158. doi: 10.18848/2474-6029/CGP/v10i01/143-158

Coronel, L. (2023). Gestión pública en la extensión universitaria para el desarrollo social: un análisis comparativo en América Latina. *Rev. Cient. Cent. Investig. Desarro.* 3, 109–125. doi: 10.23857/pc.v9i6.7307

Creswell, J. W., and Poth, C. N. (2018). *Qualitative inquiry and research design: Choosing among five approaches*. 4th Edn. California: Sage Publications.

Denzin, N. K., and Lincoln, Y. S. (2018). *The SAGE handbook of qualitative research*. 5th Edn. California: Sage Publications.

Díaz Herrera, C. (2018). Investigación cualitativa y análisis de contenido temático. Orientación intelectual de revista Universum. *Rev. Gen. Inf. Doc.* 28, 119–142. doi: 10.5209/RGID.60813

Esquivel, R. I. (2019). An econometric model to predict expected performance of Panamanian universities in international higher education rankings. *7th International Engineering, Sciences and Technology Conference (IESTEC)*. Panama: IEEE. doi: 10.1109/IESTEC46403.2019.00065

Feria Avila, H., Matilla González, M., and Mantecón Licea, S. (2019). La triangulación metodológica como método de la investigación científica. Apuntes para una

conceptualización. *Didáctica Y Educación ISSN 2224–2643* 10, 137–146. Available online at: https://dialnet.unirioja.es/descarga/articulo/7248603.pdf (Accessed February 18, 2026).

Finlay, L. (2002). Negotiating the swamp: the opportunity and challenge of reflexivity in research practice. *Qual. Res.* 2, 209–230. doi: 10.1177/146879410200200205

Flick, U. (2018). *An introduction to qualitative research. 6th* Edn. California: Sage Publications.

Forni, P., and De Grande, P. (2019). Triangulación y métodos mixtos en las ciencias sociales contemporáneas. *Rev. Mex. Sociol.* 82. doi: 10.22201/iis.01882503p.2020.1.58064

Fusch, P., Fusch, G. E., and Ness, L. R. (2018). Denzin's paradigm shift: revisiting triangulation in qualitative research. *J. Sustain. Soc. Change* 10. doi: 10.5590/JOSC.2018.10.1.02

García-Arce, J. G., Pérez- Ramírez, C. A., and Gutiérrez Barba, B. E. (2021). Objetivos de Desarrollo Sustentable y funciones sustantivas en las Instituciones de Educación Superior. *Actual. Investig. Educ.* 21, 1–34. doi: 10.15517/aie.v21i3.48160

García de Fanelli, A. M. (2019). El financiamiento de la educación superior en América Latina. *Prop. Educ.* 2, 111–126. Available online at: http://repositorio.cedes.org/handle/123456789/4597 (Accessed February 5, 2026).

Greenhalgh, T., Jackson, C., Shaw, S., and Janamian, T. (2021). Achieving research impact through co-creation in community-based health services: literature review and case study. *Milbank Q.* 94, 392–429. doi: 10.1111/1468-0009.12197

Gutiérrez-Sánchez, G., Álvarez-Muñoz, P., Galindo-Villardón, P., and Vicente-Galindo, P. (2025). Scientific collaboration and sustainable development: a bibliometric analysis of the Andean region, Panama, and Spain. *Publica* 13:10. doi: 10.3390/publications13010010

Huete-Perez, J. A., and Salvatierra, N. (2025). Assessing biomedical research capacities in selected countries of Latin America: challenges, opportunities, and recommendations. *Front. Res. Metr. Anal.* 10:1594303. doi: 10.3389/frma.2025.1594303

Instituto Nacional de Estadística y Censo. (2025). Education Publications, year 2023. Available online at: https://www.inec.gob.pa/publicaciones/Default3.aspx?ID_PUBLICACION=1313&ID_CATEGORIA=6&ID_SUBCATEGORIA=41 (Accessed February 18, 2026).

International Institute for Higher Education in Latin America and the Caribbean. (2024). Contribution of universities to the national research systems in Latin America and the Caribbean. UNESCO. Available online at: https://www.iesalc.unesco.org/en/articles/contribution-universities-national-research-systems-latin-america-and-caribbean

Kezar, A. J., and Holcombe, E. M. (2017). Shared leadership in higher education: important lessons from research and practice. Washington, DC: American Council on Education.

Korstjens, I., and Moser, A. (2017). Practical guidance to qualitative research. Part 4: trustworthiness and publishing. *Eur. J. General Practice* 24, 120–124. doi: 10.1080/13814788.2017.1375092

Krylova, P., Harmacek, J., and Htitich, M. (2025). Social progress index 1990–2020: measuring societal wellbeing over 31 years. *Int. J. Soc. Res. Methodol.* 29, –122. doi: 10.1080/13645579.2025.2486264

Kvale, S., and Brinkmann, S. (2015). *Interviews: Learning the craft of qualitative research interviewing. 3rd* Edn. California: Sage Publications.

León, M. (2020). "Exploring university engagement through an international Lens: the case of Extensión Universitaria in Panama" in *Re-envisioning higher education's public Mission: Global perspectives* (Cham: Springer International Publishing), 141–167.

Lincoln, Y. S., and Guba, E. G. (1985). *Naturalistic inquiry*. Thousand Oaks: Sage Publications, 316.

Mendigutxia, A. (2024). Contribución de las universidades a los sistemas nacionales de investigación en América Latina y el Caribe. UNESCO IESALC, Serie Análisis de Políticas de Es. Available online at: https://unesdoc.unesco.org/ark:/48223/pf0000392037 (Accessed February 18, 2026).

Mendoza, E., Barsallo, G., Ordoñez, C., and Bula, R. (2025). Validación del instrumento (ENACT) que evalúa el perfil académico de docentes y factores motivacionales hacia la investigación. *Rev. Conrado*, 21. Available online at: https://conrado.ucf.edu.cu/index.php/conrado/article/view/4228 (Accessed December 20, 2025).

Merriam, S. B., and Tisdell, E. J. (2016). *Qualitative research: A guide to design and implementation*. 4th Edn. San Francisco, California: Jossey-Bass.

Meyer, J. W., and Rowan, B. (1977). Institutionalized organizations: formal structure as myth and ceremony. Am. J. Sociol. 83, 340–363. doi: 10.1086/226550

Miller, N., Mapp, U., Bieberach, R., and López-Alfano, O. (2023). Cultura investigativa en las universidades panameñas: Realidad o ficción. XIX Congreso Nacional de Ciencia y Tecnología - APANAC 2023. doi: 10.33412/apanac.2023.3964

Moraima Campos, M., and Auxiliadora Mújica, L. (2008). El Análisis De Contenido: Una forma de abordaje metodológico. *Laurus* 14, 129–144. Available online at: https://www.redalyc.org/pdf/761/Resumenes/Resumen_76111892008_1.pdf (Accessed February 5, 2026).

Nevache, C. (2019). Visión para la ciencia, la tecnología y la innovación 2019–2024. Available online at: https://www.senacyt.gob.pa/wp-content/uploads/2018/12/quinquenio-para-la-ciencia-senacyt.pdf (Accessed February 18, 2026).

Organización de Estados Iberoamericanos para la Educación, la Ciencia y la Cultura and UNESCO. (2025). Indicadores comparativos. En El Estado de la Ciencia: Principales Indicadores de Ciencia y Tecnología 2025 (pp. 103–132). OEI. Available online at: https://oei.int/wp-content/uploads/2025/11/el-estado-de-la-ciencia-2025-web-final-terminado.pdf

Palacios, E. (2023). Barreras que enfrentan las Universidades para la transferencia de la investigación a las políticas públicas. *J. De Cienc. Soc.* 2, 69–80. doi: 10.18682/jcs.v2i21.9955

Pascual Del Rosario, G. (2024). La educacion superior y el indice de progreso social en panamá. *Cuad. De Coyuntura* 2, 60–77.

Patton, M. Q. (2015). *Qualitative research & evaluation methods: Integrating theory and practice*. 4th Edn. Thousand Oaks, California: Sage Publications.

Pereira-Medina, M. A., Gómez, C., Pinto-Martínez, M. A., Ruiz-Ochoa, M., and Meneses-Jácome, A. (2016). Transformación e innovación de la cultura organizacional en instituciones de educación superior. Estado del arte y bases conceptuales. *Revista Docencia Universitaria*, 17, 43–54. Available online at: https://revistas.uis.edu.co/index.php/revistadocencia/article/view/6774

Pfeffer, J., and Salancik, G. R. (2015). *External control of organizations: A resource dependence perspective*. Stanford, California: Stanford Business Books.

Ryan, R. M., and Deci, E. L. (2020). Intrinsic and extrinsic motivation from a self-determination theory perspective: definitions, theory, practices, and future directions. *Contemp. Educ. Psychol.* 61:101860. doi: 10.1016/j.cedpsych.2020.101860

Sánchez-Galán, E. A. (2020). La investigación universitaria: El caso de la Facultad de Ciencias Agropecuarias de la Universidad de Panamá (2007-2018). *Acción Reflex. Educ.* 45, 1–27. doi: 10.48204/j.are.n45a1

Santa Cruz Terán, F. F., Obando Peralta, E. C., Reyes Pastor, G. E., and Rodríguez-Balcázar, S. C. (2022). Investigación cualitativa: una mirada a su validación desde la perspectiva de los métodos de triangulación. Revista De Filosofía. 39, 59–72. doi: 10.5281/zenodo.6663103

Schlunegger, M. C., Zumstein-Shaha, M., and Palm, R. (2024). Methodologic and data-analysis triangulation in case studies: a scoping review. *West. J. Nurs. Res.* 46. doi: 10.1177/01939459241263011

Senacyt. (2024). Informe de indicadores de actividades de ciencia, tecnología e innovación. Observatorio Nacional de Ciencia, Tecnología e Innovación. Available online at: https://observatorio.senacyt.gob.pa/wp-content/uploads/2024/12/Informe-de-indicadores-ACT_dic_16.pdf

Soto, M. (2020). Perspectivas de la investigación en el mundo: Caso Panamá. *Rev. Global Negot.* 3, 181–183. doi: 10.53485/rgn.v3i3.147

Svenson, N. A., Bernett, G., de Gracia, G., De Leon, N., Psychoyos, D., and Rao, K. S. J. (2018). Panamanian start-ups that propel research and higher learning. *Investigación y Pensamiento Crítico* 6, 68–79. doi: 10.37387/ipc.v6i3.96

Szulawski, M., Kaźmierczak, I., and Prusik, M. (2021). Is self-determination good for your effectiveness? A study of factors which influence performance within self-determination theory. *PLoS One* 16:e0256558. doi: 10.1371/journal.pone.0256558

Tejedor De León, A. (2021). JIC: scientific initiation day to promote university research from early stages. *J. High. Educ. Theory Pract.* 21, 107–108. doi: 10.33423/jhetp.v21i11.4668

United Nations Economic Commission for Latin America and the Caribbean (ECLAC). (2023). Science, technology and innovation for sustainable development in Latin America and the Caribbean. Available online at: https://repositorio.cepal.org/handle/11362/48683

Valdés-Montecinos, M., and Ganga-Contreras, F. (2021). Gobernanza universitaria: Aproximaciones teóricas de los grupos de interés en Instituciones de Educación Superior. *Rev. Cienc. Soc.* 27, 441–459. doi: 10.31876/rcs.v27i3.36781

Villadiego Sánchez, L. I. (2020). La cultura organizacional en las instituciones de educación superior: Elemento clave para fortalecer el aprendizaje institucional. *Revista Dialogus*, 4, 29–36. diciembre 2020–mayo 2021. doi: 10.37594/dialogus.v1i6.564

Villalba Benítez, E. F. (2017). Desafíos de la gestión universitaria: Reflexiones en torno a las prácticas y tendencias en Paraguay. *Rev. Argent. Educ. Super.* 15, 36–53. Available online at: https://dialnet.unirioja.es/descarga/articulo/6530578.pdf